

Deep Convolutional Factor Prediction with Explainable AI for Enhanced Investment Strategy

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Abstract

Factor models are widely used to explain stock market behavior and to forecast returns in financial applications. However, traditional regression-based approaches struggle to capture nonlinear dynamics and exhibit vulnerability to noise when directly predicting asset returns in high-dimensional settings. To overcome these limitations, this paper proposes ConvFactorNet—a deep learning-based factor prediction system designed to extract latent common factors from market data and integrate them with asset-specific beta coefficients for improved return forecasting. Empirical analysis utilizing Korean stock market data from 2015 to 2023 reveals that a long–short strategy based on ConvFactorNet achieved a Compound Annual Growth Rate (CAGR) of 21.67% and a Sharpe ratio of 3.78, substantially outperforming benchmarks. To enhance interpretability, gradient-weighted class activation mapping (Grad-CAM) was incorporated, which reveals the importance of macroeconomic indicators for predictive decisions. Retraining the model using only these critical features further improved cumulative return and Sharpe ratio, highlighting the value of explainable AI in feature selection and model reliability. The contributions of this paper are threefold: first, ConvFactorNet overcomes key shortcomings of conventional factor models; second, empirical evidence demonstrates its superior and robust investment performance; and third, Grad-CAM not only improves interpretability but also strengthens predictive accuracy and practical deployment in intelligent trading applications.

Keywords:

Deep learning; Convolutional neural networks; Factor modeling; Stock return prediction; Explainable AI

1. Introduction

Return prediction plays a central role in various domains of financial markets, including asset allocation, risk management, and the development of intelligent investment strategies. Traditionally, expert systems for return prediction have relied on the factor model, which employs factors—variables that explain the common variation in asset returns. These factors serve as economically interpretable sources of risk or latent variables extracted through statistical techniques. The well-known Fama-French Three-Factor Model (Fama & French, 1993), for instance, incorporates market risk, the size effect (SMB: Small Minus Big), and the value effect (HML: High Minus Low), providing a robust quantitative framework for capturing investor risk exposures and extending asset pricing theory. However, most conventional factor models are built on the assumption of linear interactions, which severely constrain their capacity to capture the complex and nonlinear behavior that characterizes modern financial markets. In practice, macroeconomic indicators (e.g., interest rates, inflation), firm-level financial metrics (e.g., profitability ratios, debt ratios), and market dynamics (e.g., stock price volatility, index returns) interact in deeply nonlinear and dynamic ways. Mandelbrot (1963) showed that asset returns deviate from normality, exhibiting fat tails and self-similarity, and Cont (2001) highlighted features such as non-stationarity, nonlinear dependencies, and volatility clustering—properties that are inadequately modeled by traditional statistical approaches.

To address these challenges, advanced deep learning methods have emerged as viable alternatives, capable of modeling high-dimensional and nonlinear structures from financial data (LeCun et al., 2015). Deep neural networks have demonstrated notable success in stock price prediction (Fischer & Krauss, 2018), risk management (Mai et al., 2019), and portfolio optimization (Heaton et al., 2017), frequently surpassing the predictive accuracy of traditional models. Nevertheless, the adoption of deep learning within expert financial systems remains challenging due to its “black-box” nature (Doshi-Velez & Kim, 2017), which can inhibit interpretability and hinder regulatory compliance. Given that practical financial systems must prioritize both predictive accuracy and transparency, models must balance predictive power with interpretability. This motivates the development of factor prediction frameworks that combine deep learning’s ability to capture nonlinearities with the enhanced interpretability provided by explainable artificial intelligence (XAI) techniques that enhance transparency.

To address these challenges, this study introduces ConvFactorNet, a deep learning-based factor prediction framework with three main contributions for expert financial applications:

1. **Integration of CNNs with Factor Models:** ConvFactorNet combines traditional factor modeling with Convolutional Neural Networks, training on multi-dimensional financial data to learn latent factors that capture complex nonlinear patterns inherent in financial markets.
2. **Novel Return Prediction Mechanism:** The learned factors are combined with asset-specific beta coefficients to generate future return predictions, directly linking return forecasting to interpretable factor structures and enhancing the transparency of the predictive system.
3. **Interpretability via Grad-CAM:** To overcome the black-box limitation of deep learning, ConvFactorNet incorporates Gradient-weighted Class Activation Mapping (Grad-CAM), enabling visualization and quantification of feature importance during factor learning—thereby supporting both explainability and trust in the expert system.

This dual emphasis on predictive performance and interpretability ensures that ConvFactorNet delivers enhanced return forecasting and demonstrates strong practical value for developing robust investment strategies in real-world financial markets.

The remainder of this paper is organized as follows: Section 2 reviews the related literature on portfolio factor models, deep learning applications in finance, and explainable AI methods. Section 3 introduces the proposed model, detailing the ConvFactorNet architecture, its training procedure, the factor prediction process, and Grad-CAM integration. Section 4 presents the dataset, experimental procedures, and evaluation results in terms of predictive accuracy and interpretability. Finally, Section 5 concludes the paper by summarizing major contributions, discussing implications for expert system deployment, and suggesting future research directions.

2. Literature Review

2.1. Portfolio Factor Models

Portfolio factor models are fundamental tools in asset pricing and portfolio management, aiming to explain expected returns and capture systematic variation among assets in financial expert systems. The Capital Asset Pricing Model (CAPM; Sharpe, 1964) pioneered the use of beta coefficients to quantify systematic risk as a determinant of expected returns. Although groundbreaking, its reliance on a single-factor specification limited its effectiveness in representing

the diversity and complexity of financial markets.

To address this limitation, Ross (1976) proposed the Arbitrage Pricing Theory (APT), a multi-factor framework incorporating a broader range of economic risk sources. Building on these advancements, the Fama and French (1993) introduced the three-factor model by adding size (SMB) and value (HML) factors to market risk, significantly improving explanatory power in empirical applications. Subsequent research has further enriched the factor modeling domain: Chen et al. (1986) incorporated macroeconomic variables; Carhart (1997) added momentum; Fama and French (2015) proposed a five-factor model including profitability (RMW) and investment (CMA); Hou et al. (2015) applied q-theory to profitability and investment.

Although these models substantially advanced the quantitative foundation for asset pricing and return prediction, most remain limited to linear interaction structures. As a result, capturing the full complexity and the nonlinear relationships present in real-world financial markets remains a key challenge for current expert systems and motivates the exploration of more flexible, data-driven approaches.

2.2. Deep Learning in Factor Modeling and Financial Prediction

Recent studies highlight the potential of machine learning and deep learning to extend traditional factor modeling in expert financial systems. Kelly, Pruitt, and Su (2019) proposed Instrumented Principal Component Analysis (IPCA) to flexibly capture time-varying risk exposures. Gu, Kelly, and Xiu (2020) showed that advanced nonlinear models such as Random Forest, Boosting, and deep learning outperform classical linear regression for asset return prediction. In further extension, Gu, Kelly, and Xiu (2021) employed Conditional Autoencoders to learn nonlinear mappings between factors and loadings. Freyberger, Neuhierl, and Weber (2020) used LASSO techniques to identify characteristics with independent predictive power, enhancing factor parsimony.

Concurrently, deep learning architectures have gained prominence in diverse financial prediction applications. Fischer and Krauss (2018) demonstrated the superiority of Long Short-Term Memory (LSTM) networks for stock forecasting; Hoseinzade and Haratizadeh (2019), Mehtab and Sen (2020), and Chen and Huang (2021) applied convolutional neural networks (CNNs) to financial data with strong results; Transformer-based models have also shown promise (Wang et al., 2022; Zhang et al., 2022). Collectively, these studies demonstrate that deep learning methods adeptly uncover nonlinear structures and improve predictive outcomes relative to traditional approaches. Despite these advances, most works focus primarily on predictive performance, with the interpretability of deep learning-based factor models remaining a notable open challenge for expert system deployment in finance.

2.3. Explainable AI in Finance

Explainable AI (XAI) has emerged as a critical response to the black-box challenges of deep learning. Transparent models (Rudin, 2019) and post-hoc methods such as LIME (Ribeiro et al., 2016), SHAP (Lundberg & Lee, 2017), and DeepLIFT (Shrikumar et al., 2017) provide tools for interpretability. In computer vision, methods like Class Activation Mapping (CAM; Zhou et al., 2016) and its extension Grad-CAM (Selvaraju et al., 2017) visualize regions influencing CNN predictions, offering intuitive interpretability.

The need for XAI is particularly heightened in finance, given the sector's strong requirements for trust, transparency, and regulatory compliance. XAI techniques have been applied across diverse financial applications including risk management (Sigrist & Hirnschall, 2019; Adams & Hagrass, 2020), portfolio optimization (Zhu et al., 2020; Guan & Liu, 2021; Babaei et al., 2022), fraud detection (Kute et al., 2021; Lin & Gao, 2022), and stock prediction (Ito et al., 2021; Kim et al., 2023). These works demonstrate that XAI can enhance transparency and reliability in financial applications. Nevertheless, their systematic adoption in factor modeling for financial prediction remains emerging, suggesting substantial potential for further methodological refinement and industry impact.

3. Method

Traditional factor models describe stock returns as a linear function of common factors:

$$\mathbf{R} = \mathbf{F} \cdot \mathbf{B} + \boldsymbol{\epsilon} \quad (1)$$

where $\mathbf{R} \in \mathbb{R}^{t \times n}$ denotes the matrix of returns for n stocks over t periods, $\mathbf{F} \in \mathbb{R}^{t \times m'}$ represents the values of m' factors across t periods, $\mathbf{B} \in \mathbb{R}^{m' \times n}$ is the stock-specific beta matrix, and $\boldsymbol{\epsilon} \in \mathbb{R}^{t \times n}$ captures idiosyncratic residuals. Here, t denotes the time length, m' is the number of factors, and n represents the number of stocks. Based on historical return data, Ordinary Least Squares (OLS) is typically applied to estimate the betas in $\mathbf{B}$. Each beta reflects a stock's

sensitivity to the corresponding factor.

Despite their foundational role, this approach faces notable limitations. Firstly, factor selection often depends on subjective human judgment based in theoretical assumptions and empirical precedents – for example, market, size, and value factors. This subjectivity can limit model robustness under evolving market conditions or new financial environments, potentially impairing the model's generalizability. Secondly, traditional factor models assume linear relationships between input and output variables, restricting their ability to capture nonlinear correlations and the complexity of high-dimensional financial data. As a result, they are limited in reflecting the intricate interactions and nonlinear structures observed in real financial markets, which can lead to reduced predictive accuracy.

To overcome these limitations, this study proposes ConvFactorNet, a novel deep learning-based factor prediction framework. ConvFactorNet autonomously learns complex, nonlinear patterns and latent factor representations directly from multi-dimensional financial data, reducing reliance on subjective factor selection and enhancing adaptability to dynamic market conditions. This approach seeks to improve both predictive accuracy and practical applicability in financial expert systems.

3.1. ConvFactorNet

ConvFactorNet leverages historical financial data to estimate stock-specific beta coefficients $\hat{\mathbf{B}}$ during a designated beta estimation period using OLS. Subsequently, it receives data from the target prediction period to forecast future factor values $\hat{\mathbf{F}}$. The predicted future returns are obtained by multiplying the predicted factors $\hat{\mathbf{F}}$ with the estimated betas $\hat{\mathbf{B}}$, which is expressed by the following equation:

$$\hat{\mathbf{R}} = \hat{\mathbf{F}} \cdot \hat{\mathbf{B}} \quad (2)$$

Here, $\hat{\mathbf{R}}$ denotes the predicted future returns, $\hat{\mathbf{F}}$ represents the predicted factor values for the future period obtained from ConvFactorNet, and $\hat{\mathbf{B}}$ refers to the stock-specific beta coefficients estimated from historical return data using OLS. This approach allows the deep learning model to adapt to changing market conditions and to capture the full complexity and effectively learn nonlinear relationships and correlations within the input financial data.

The proposed model input consists of feature data $\mathbf{X} \in \mathbb{R}^{n \times m}$, representing m features for n stocks at a specific time point, to predict m' factor values $\hat{\mathbf{F}}_t \in \mathbb{R}^{m'}$. This process is expressed by the following equation:

$$\hat{\mathbf{F}}_t = f(\mathbf{X}; \theta) \quad (3)$$

Here, f denotes the ConvFactorNet model, and θ represents the learned parameters of ConvFactorNet. These predicted factor vectors for each time point ($\hat{\mathbf{F}}_t$) are then stacked to form the future factor matrix $\hat{\mathbf{F}} = [\hat{\mathbf{F}}_1, \hat{\mathbf{F}}_2, \dots, \hat{\mathbf{F}}_t]^T \in \mathbb{R}^{t \times m'}$, which is subsequently used in the return prediction in Equation (2).

Figure 1 is here

The model's architecture is designed to process input financial data $\mathbf{X} \in \mathbb{R}^{n \times m}$ and learn complex correlations among stocks and features through its CNN-based structure. This architecture enables the extraction of factor values that effectively represent prevailing market conditions at each specific time point. The input data sequentially passes through three blocks, each comprising Conv2D, BatchNorm2D, ReLU, MaxPool2D layers, with the number of channels progressively increasing to enhance feature abstraction. The final convolutional layer reduces the channels dimension to the number of factors m' , after which Global Average Pooling (GAP) is applied to output the final factor values for each time point. This design efficiently captures nonlinear dependencies and hierarchical feature interactions within high-dimensional financial data.

Table 1 is here

3.2. Model Training

ConvFactorNet is trained through a four-step process to optimize stock return prediction. Each step is designed to learn meaningful patterns from financial data and estimate factor values and beta coefficients that can be effectively used for stock return prediction.

Training Step 1: Factor Prediction on Batch Data

In the first stage of training, the model receives batch-wise input from the beta estimation dataset to predict factor values at each time point. The input data is structured in the shape $(b \times n \times m)$, where b denotes the batch size corresponding to time steps or dates, n is the number of stocks, and m is the number of features per stock. ConvFactorNet takes the batch data as input and predicts the factor values corresponding to each time point. The dimensionality of the predicted factor values is set to m' , which can vary depending on the experimental configuration. The predicted factors are represented as $\hat{\mathbf{F}}_{batch} \in \mathbb{R}^{b \times m'}$. This step enables the model to learn latent market factors that vary over time.

Training Step 2: Betas Estimation via OLS on Batch Data

In this step, stock-specific beta values are estimated based on the factor values from Step 1. By applying OLS to the predicted factor values $\hat{\mathbf{F}}_{batch}$ from Step 1 and the actual return data $\mathbf{R}_{batch}$ within the batch, the beta values for each stock can be estimated. The estimated beta matrix is expressed as $\hat{\mathbf{B}}_{batch} \in \mathbb{R}^{m' \times n}$, which represents each stock's sensitivity to the latent factors over the batch period and forms an essential component for subsequent return prediction. The estimation is calculated using the following equation:

$$\hat{\mathbf{B}}_{batch} = (\hat{\mathbf{F}}_{batch}^T \hat{\mathbf{F}}_{batch} + \lambda \mathbf{I})^{-1} \hat{\mathbf{F}}_{batch}^T \mathbf{R}_{batch} \quad (4)$$

Here, λ is a regularization term introduced for numerical stability. In this study, the regularization value is set to $\lambda = 1 \times 10^{-4}$.

Training Step 3: Returns Prediction on Batch Data

Predicted returns ($\hat{\mathbf{R}}_{batch}$) for each batch are calculated by multiplying the estimated beta matrix ($\hat{\mathbf{B}}_{batch}$) with the predicted factor values ($\hat{\mathbf{F}}_{batch}$), expressed as:

$$\hat{\mathbf{R}}_{batch} = \hat{\mathbf{F}}_{batch} \cdot \hat{\mathbf{B}}_{batch} \quad (5)$$

Training Step 4: Loss Calculation and Backpropagation

Finally, the difference between the predicted and actual returns is used to compute the loss. This study uses Root Mean Square Error (RMSE) loss function:

$$\mathcal{L}_{RMSE} = \sqrt{\frac{1}{b \cdot n} \sum_{i=1}^b \sum_{j=1}^n (R_{i,j} - \hat{R}_{i,j})^2} \quad (6)$$

where $R_{i,j}$ denotes the actual return of the j -th stock at the i -th time point in the batch, and $\hat{R}_{i,j}$ is the corresponding predicted return obtained using ConvFactorNet. Here, b and n represent the batch size and the number of stocks, respectively. The loss value reflects the model's predictive accuracy.

The model parameters are then updated using the Adam optimizer and the backpropagation algorithm to minimize this loss iteratively. As a result, ConvFactorNet learns to generate factor values $\hat{\mathbf{F}}$ that effectively explain stock returns $\mathbf{R}$ at a given time, using the input financial feature data $\mathbf{X}$. This training process enables the model to learn complex patterns and interactions embedded in the data and to extract factors that provide a compact and effective representation of high-dimensional financial information at each time point, and progressively improving prediction performance.

3.3. Target Data Prediction

The objective of this study is to predict stock-specific returns at future time points using the trained ConvFactorNet model in conjunction with historical market data. The prediction process consists of three key steps.

Prediction Step 1: Betas Estimation via OLS on Beta Estimation Data

First, the trained ConvFactorNet model is used to predict factor values for each time point within the beta estimation period, which is the historical window preceding the target date. The input for this step is the beta estimation data, and the resulting predicted factor values $\hat{\mathbf{F}}_{past}$ represent a low-dimensional embedding that effectively captures the market conditions and data patterns during the beta estimation period.

Next, these predicted factors $\hat{\mathbf{F}}_{past}$ are combined with the actual observed returns $\mathbf{R}_{past}$ from the beta estimation period to estimate stock-specific beta values $\hat{\mathbf{B}}_{past}$ using Ordinary Least Squares (OLS). This process is expressed as:

$$\hat{\mathbf{B}}_{past} = (\hat{\mathbf{F}}_{past}^T \hat{\mathbf{F}}_{past} + \lambda I)^{-1} \hat{\mathbf{F}}_{past}^T \mathbf{R}_{past} \quad (7)$$

Here, $\hat{\mathbf{B}}_{past} \in \mathbb{R}^{m' \times n}$ and $\hat{\mathbf{F}}_{past} \in \mathbb{R}^{t \times m'}$, where t denotes the length of the beta estimation period and m' is the number of factors—both of which are hyperparameters predetermined by experimental design, ensuring flexibility for varying forecasting horizons and model complexity. In the empirical analysis, various values of these hyperparameters are tested to compare performance outcomes. This process yields a matrix of stock-specific betas that reflect each asset's recent sensitivities to the discovered factors, forming the foundation for robust target-date return prediction.

Prediction Step 2: Factors Prediction on Target Data

This step is where the model generates the current factor exposure needed for the target return forecast. At this stage, the most up-to-date feature matrix for all stocks ($\mathbf{X}_{target}$) is used as input to the trained ConvFactorNet model. Through the ConvFactorNet model, this input is mapped to a vector of predicted factor values ($\mathbf{F}_{target}$) for the target future time point. The predicted factor values $\hat{\mathbf{F}}_{target}$ generated in this step are represented as follows:

$$\hat{\mathbf{F}}_{target} = f(\mathbf{X}_{target}; \theta) = [\hat{F}_{target}^1, \hat{F}_{target}^2, \dots, \hat{F}_{target}^{m'}] \quad (8)$$

Here, $\mathbf{X}_{target} \in \mathbb{R}^{1 \times n \times m}$ is the feature matrix at the target time point, and $\hat{\mathbf{F}}_{target} \in \mathbb{R}^{1 \times m'}$ is the predicted factor vector. f denotes the trained factor prediction model (ConvFactorNet), and θ refers to the model's learned parameters. Each $\hat{F}_{target}^i$ represents the predicted value of the i -th factor, and the model outputs a total of m' factors. The total number of factors generated by the model is controlled as a hyperparameter defined in the experimental setup. This process ensures that the return forecasts for the future date are specifically informed by the latest market and company conditions rather than relying solely on historical averages or static factors. The resulting $\hat{\mathbf{F}}_{target}$ thus forms the key dynamic ingredient for the subsequent step, where predicted returns are computed through the combination with stock-specific betas.

Prediction Step 3: Returns Prediction on Target Data

The final prediction step involves predicting future returns by combining the factor values predicted at the target time point with the stock-specific betas estimated during the beta estimation period.

$$\hat{\mathbf{R}}_{target} = \hat{\mathbf{F}}_{target} \cdot \hat{\mathbf{B}}_{past} \quad (9)$$

Here, $\hat{\mathbf{R}}_{target} \in \mathbb{R}^{1 \times n}$ is the predicted return matrix for n stocks over the forecast horizon, which is set as a hyperparameter in the experiment. This study compares results for 20 trading days (1 month), 60 trading days (3 months), and 120 trading days (6 months). This prediction captures the expected return for each individual stock at the target date, integrating dynamic factor exposures with empirically derived risk sensitivities.

The final predicted return values serve as direct inputs for investment strategy development and portfolio optimization. The model's predictive performance is evaluated by comparing the predicted returns with actual market outcomes. This study assesses how effectively the model responds to real market conditions and based on this evaluation verifies the practical utility of the proposed approach. This validation confirms the capability of ConvFactorNet to adaptively

and effectively model the evolving financial environment for superior predictive performance.

3.4. Grad-CAM Application

Grad-CAM improves upon its predecessor, Class Activation Mapping (CAM), by leveraging the gradients of the final convolutional layer. This approach overcomes the limitations of CAM, which requires a Global Average Pooling (GAP) layer that can restrict model architecture and degrade performance. By using gradient information, Grad-CAM generates a localization map, or heatmap, that highlights the most important input regions for a particular prediction, making it a more flexible and effective tool for interpreting a wide range of CNN architectures. In this study, we apply Grad-CAM to identify the input features that most significantly influence ConvFactorNet's factor value predictions. This visualization technique highlights the model's attention to specific features and stocks, thereby enhancing its interpretability and trust in the model's predictions. This helps stakeholders understand the decision-making process of the deep learning model in financial forecasting contexts.

3.4.1. Grad-CAM Calculation

Grad-CAM is applied to the final convolutional layer of ConvFactorNet, specifically Conv2d-13, to calculate the importance of different feature maps for each predicted factor. For each predicted factor c (where c ranges from 1 to m'), the importance weights a_k^c are first calculated. These weights represent the importance of the k -th feature map for predicting the factor c . The calculation uses the gradients of the score for factor c (denoted as $\hat{F}_{target}^c$) with respect to the activation maps A^k of the final convolutional layer.

$$a_k^c = \frac{1}{H \times W} \sum_i \sum_j \frac{\partial \hat{F}_{target}^c}{\partial A_{i,j}^k} \quad (10)$$

Here, $H \times W$ is the total number of pixels in the feature map, and $A_{i,j}^k$ refers to the activation at spatial location (i,j) . These weights are then used to compute the heatmap for factor c , L^c , formed as a weighted sum of the forward activation maps followed by a ReLU activation:

$$L^c = \text{ReLU} \left(\sum_k a_k^c A^k \right) \quad (11)$$

To create a single comprehensive visualization across all factors, this study computes a total heatmap, L^{total} , by taking a weighted sum of the individual factor maps. The weights are the absolute values of the predicted factor scores, emphasizing factors with stronger signals.

$$L^{total} = \sum_{c=1}^{m'} |\hat{F}_{target}^c| \cdot L^c \quad (12)$$

This combined map is then resized to match the dimensions of the original input feature matrix using linear interpolation. This resizing aligns the Grad-CAM heatmap with the shape of the input feature matrix, enabling direct mapping and interpretation of which features were considered most important by the model during prediction.

3.4.2. Grad-CAM Visualization

The Grad-CAM results are visualized as heatmaps that enhance interpretability by showing which features and stocks the model focuses on when predicting factors. The visualization is carefully ordered:

- Y-axis: Individual stocks are sorted by market capitalization, from Big-cap to Small-cap.
- X-axis: The 110 input features are categorized into Technical, Financial, Macro, and Market Indicators, with vertical lines as separators.

An example of the resulting Grad-CAM visualization is presented below, illustrating the relative importance of input features as perceived by the model during factor prediction. In the heatmap visualization, red regions indicate the variables that the model considered most important when predicting the factors, while blue regions represent those that were considered relatively less important.

Figure 4 is here

Based on this visual analysis, the study investigates how ConvFactorNet differentially reflects the structural characteristics of financial data. In particular, when certain feature categories or groups of stocks exhibit prominently high Grad-CAM activation values, it indicates that the model perceives those variables as key components in factor estimation. These findings contribute significantly to understanding the internal logic of deep learning-based factor models, improving transparency by showing the reasoning behind their predictions and guiding further model refinement or feature selection.

4. Experiments

4.1. Performance Evaluation of ConvFactorNet

4.1.1. Data Collection and Preprocessing

To evaluate the performance of the proposed ConvFactorNet model, this study conducts an empirical analysis using daily stock data from companies listed on the KOSPI and KOSDAQ markets in South Korea. The dataset spans from January 2010 to December 2023. To ensure data quality, specific industries such as SPACs and financial firms were excluded, and only companies with a market capitalization exceeding KRW 100 billion were included.

A comprehensive set of 110 features was utilized, categorized into four distinct groups:

- Technical Indicators: Derived from historical price and trading volume data, with price-based metrics normalized by closing price and volume-based metrics scaled by trading volume.
- Financial Indicators: Extracted from corporate financial statements, with a four-month reporting lag applied to prevent look-ahead bias and ensure data reflects publicly available information at prediction time.
- Macro Indicators: Key macroeconomic variables converted from monthly to daily frequency using forward-fill interpolation.
- Market Indicators: Historical returns from major global market indices such as the KOSPI, NASDAQ, and S&P 500 across various time horizons.

Following feature-specific preprocessing, all 110 features were collectively normalized using a MinMaxScaler. The model's target variable is the future stock return, evaluated over three distinct forecasting periods: 20 trading days (1 month), 60 trading days (3 months), and 120 trading days (6 months). A detailed list of all 110 features and their respective categories is provided in Appendix A.

4.1.2. Experimental setup

The experimental period spans from January 2015 to May 2023. For model evaluation, the last trading day of each month within this period was designated as a target date, resulting in a total of 101 distinct evaluation points. For each target date, the ConvFactorNet model was trained on a preceding historical data window (the lookback period) to predict cumulative stock returns over a subsequent future period (the target days). To systematically assess the model's generalization capabilities and predictive accuracy, a comprehensive hyperparameter tuning process was conducted. A total of 36 unique hyperparameter combinations were tested, focusing on three key variables:

1. Lookback Period: Duration of historical data used for model training, set to 1 year, 2 years, and 3 years to analyze the impact of data history on performance.
2. Target Days: Prediction horizon for future returns. As previously defined, this was set to 20, 60, and 120 days to evaluate short-, medium-, and long-term forecasting accuracy.
3. Number of Factors (m'): Output dimensionality of ConvFactorNet. This varied across 5, 10, 20, and 40 factors to examine how model complexity affects predictive power.

All models were trained for 300 epochs with a batch size of 64. The training objective was to minimize the Root Mean Square Error (RMSE) between the predicted and actual returns. Optimization was performed using the Adam optimizer with a learning rate of 0.001.

The implementation was carried out in Python using the PyTorch deep learning framework. All training and inference procedures were executed on an NVIDIA GeForce RTX 3090 GPU. The results of these experiments were evaluated based on two primary categories: predictive performance and investment performance, which are detailed in the subsequent sections.

4.1.3. Predictive Performance

To quantitatively assess the ConvFactorNet's predictive power, continuous cumulative return forecasts are transformed into a binary classification task. The model's objective is to predict the direction of the cumulative stock return over the forecast horizon, classifying a forecast as "up" (1) for a positive return and "down" (0) for a negative one. These classifications are then benchmarked against actual return directions. The actual cumulative return $CR_{i,t}$ for stock i over the target window is formulated as:

$$CR_{i,t} = \left(\prod_{s=1}^{\text{target days}} (1 + r_{i,t+s}) \right) - 1 \quad (13)$$

Model performance is evaluated using two primary metrics:

- Accuracy, defined as the proportion of correct predictions, can be misleading for imbalanced financial data.
- Macro F1 Score is adopted to provide a more robust evaluation by treating both upward and downward movements with equal importance.

The model's performance was evaluated across 36 distinct hyperparameter combinations, with the results sorted by accuracy as shown in Table 2. The optimal configuration was achieved with a 2-year lookback period, a 120-day (six-month) prediction horizon, and 20 factors, attaining an accuracy of 54.72% and a Macro F1 Score of 0.5002.

An analysis of the hyperparameter settings revealed several key insights. The number of factors significantly impacted performance; using excessive factors (40) caused overfitting, while too few (5) resulted in an insufficient capacity to capture market patterns. The model also demonstrated stronger performance on longer-term predictions (60 and 120 days) compared to short-term horizons (20 days). In contrast, varying the lookback period did not yield significant differences in accuracy.

In summary, the results confirm that the model's effectiveness is contingent on proper hyperparameter selection, particularly the number of factors and the prediction horizon. The model is most suitable for capturing long-term stock market movements with appropriately chosen parameters.

Table 2 is here

4.1.4. Investment Performance

To evaluate the practical applicability of ConvFactorNet, the model's investment performance was assessed through a comprehensive backtesting procedure. This process utilized a long-short strategy to form a market-neutral portfolio, allowing for an evaluation of the model's ability to generate excess returns independent of the overall market movements.

The performance of long-short portfolios was evaluated using a comprehensive set of metrics, with the results for the 36 hyperparameter configurations summarized in Table 3. The key metrics are defined as follows:

- Tertile Accuracy: Proportion of stocks for which the predicted return tercile (High, Medium, or Low) matches the realized return tercile.
- Cumulative Return: Total increase in portfolio value over the investment period.
- Compound Annual Growth Rate: Average annual growth rate across the investment horizon.
- Volatility: Annualized standard deviation of monthly returns, representing the portfolio's risk.
- Sharpe Ratio: Ratio of the excess return over risk-free rate to return volatility, measuring risk-adjusted performance. The risk-free rate was assumed to be 2%.
- Maximum Drawdown (MDD): Largest observed loss from a peak to a trough during the investment period.

The model that achieved the highest investment performance was trained on 1-year lookback window, utilized 10 factors (m'), and predicted 6-month (120-day) forward returns. Over a 101-month period, this model generated a cumulative return of 412.44% with a low volatility of 5.21%, achieving a Sharpe Ratio of 3.78. This model also recorded the highest Tertile Accuracy of 38.52%.

Analysis of hyperparameter impacts on investment performance revealed that longer prediction horizons and factor counts of 10 or 20 yielded superior investment outcomes. Models with too many ($m'=40$) or too few ($m'=5$) factors showed degraded performance, whereas the lookback period exhibited a relatively minor effect. For a deeper diagnostic analysis of the model's classification behavior, a detailed examination of the confusion matrices and an analysis of monthly returns by tercile group are provided in [Appendix B](#).

Table 3 is here

To further evaluate the effectiveness of ConvFactorNet, its performance was benchmarked against three distinct strategies. The first, the KOSPI Index, represents overall market performance, serving as a baseline market benchmark. The second, CNN-Direct, is a variant of our model that predicts stock returns directly without the intermediate factor-learning step, serving as an architectural benchmark to evaluate the benefit of factor-based modeling. The third, a traditional Small High Strategy, provides a comparison against conventional factor investing. As visualized in the cumulative return trends in Fig. 5, the ConvFactorNet strategy significantly outperformed all benchmarks across the evaluation period. Notably, compared with CNN-Direct, ConvFactorNet demonstrated significantly improved Tertile Accuracy and return metrics, evidencing the structural advantage of its factor-based approach, which enhances the model's ability to accurately rank stocks.

Furthermore, the model's clear outperformance against both the market (KOSPI) and the traditional factor strategy highlights its superior ability to capture complex, nonlinear market dynamics. This enables ConvFactorNet to deliver substantial alpha beyond what simpler linear or direct prediction models can achieve, affirming its practical value for advanced investment decision-making.

Figure 5 is here

4.2. Explainability Analysis Using Grad-CAM

After confirming predictive and investment performance of ConvFactorNet, Gradient-weighted Class Activation Mapping (Grad-CAM) was employed to systematically analyze the model's decision-making process. The primary objective of this analysis is to enhance the transparency and interpretability of the factor estimation mechanism by identifying the input features that are most influential.

Grad-CAM analysis is conducted on the final convolutional layer (Conv2d-13) of the trained ConvFactorNet. When the target data, $X_{target} \in \mathbb{R}^{1 \times n \times m}$, is processed, the model outputs m' factor values. By computing a Grad-CAM map for each of these m' factors, it is possible to quantitatively assess which input features were most critical in determining each specific factor's value. The analysis utilizes the hyperparameter configuration that demonstrated the highest predictive performance: a 1-year lookback period, 120 target days, and 10 factors (m').

For visual interpretation, the activation maps generated by Grad-CAM are rendered as heatmaps. In these visualizations, the Y-axis represents individual firms sorted by market capitalization (from large-cap to small-cap), while the X-axis displays the input features, categorized as Technical, Financial, Macro, and Market indicators. To create a comprehensive overview, the Grad-CAM maps for each individual factor are weighted by the absolute value of their corresponding factor estimates and then aggregated into a single, final output.

The analysis, exemplified by the Grad-CAM output for April 30, 2021, reveals that the model utilizes a balanced combination of features from all indicator categories. Rather than relying heavily on a single group of indicators, ConvFactorNet generates factor values through a composite use of diverse feature groups. Additionally, feature importance remained relatively stable across firms of different sizes. This suggests that the overall cross-sectional patterns in the feature data exerted a greater influence on the factor estimation process than firm-specific characteristics.

4.2.1. Grad-CAM Results Over the Entire Period

An analysis of the average Grad-CAM values across the entire target period (from January 30, 2015, to May 31, 2023) was conducted to identify the variable groups prioritized by ConvFactorNet when predicting factor values.

Figure 6 is here

Based on the average Grad-CAM scores, Macro variables consistently emerged as the most influential feature group, suggesting that macroeconomic factors play a dominant role in factor prediction relative to firm-specific characteristics. Following Macro variables, Market variables were identified as the next most important feature group. A quantitative analysis of the average Grad-CAM values by feature category, presented in Table 4, confirms this hierarchy, showing that the Macro category exhibited the highest importance, followed by the Market and Financial categories. In contrast, the Technical category showed relatively low importance, suggesting the model is primarily focused on capturing persistent, long-horizon economic and market trends, rather than short-term technical patterns.

Table 4 is here

An examination of feature importance at the individual variable level, detailed in Table 5, revealed that the Korea Industrial Production Index, US Industrial Production Index, Gold Futures, WTI Futures, and the US Treasury Bond (10-Year) recorded the highest Grad-CAM values. These results indicate that the factor model emphasizes variables that reflect broad economic trends and business cycle volatility over those linked to short-term stock price fluctuations.

Table 5 is here

Furthermore, additional detailed analyses were conducted to investigate whether these patterns of feature importance hold constant across industry sectors. [The results of these analyses are provided in Appendix C.](#) The analysis confirms the robustness of these patterns. Across sectors, ConvFactorNet relies more heavily on macroeconomic and market indicators than on firm-specific features when predicting factors. Collectively, these findings demonstrate that ConvFactorNet is better suited to modeling long-term economic dynamics, rather than reacting to short-term market volatility.

4.2.2. Grad-CAM Results by Market Regime

To examine Confectioner's adaptability across varying market environments, a regime-segmented Grad-CAM analysis was conducted. This investigation aimed to determine whether the model assigns different levels of importance to input features in response to prevailing market regimes.

Algorithm 1 is here

A market regime classification algorithm was developed based on monthly KOSPI returns, with the full procedure detailed in Algorithm 1. The process involves smoothing monthly returns with a centered three-month moving average to reduce short-term volatility. Regime labels—bull, bear, or flat—are then assigned based on threshold rules. To ensure regime continuity and prevent fragmentation from isolated market fluctuations, a label smoothing mechanism was applied. Using these generated regime labels, all model prediction periods were grouped into their corresponding market phases to compute and compare Grad-CAM values separately for each regime.

Figure 7 is here

The classification results are illustrated in Fig. 7, which displays the KOSPI index with background colors indicating the identified market regimes. The segmentation accurately captures major macroeconomic episodes, such as the downturn during the COVID-19 pandemic and the subsequent rebound, as well as prolonged downtrends like the 2018–2019 bear market. This framework forms the basis for further analysis to identify whether ConvFactorNet's factor attention varies under distinct market conditions.

Table 6 is here

As summarized in Table 6, the classification of the 101-month period yielded 46 bull market months, 32 bear market months, and 23 flat market months. These results confirm that the regime identification algorithm provides a consistent and interpretable segmentation of market states, forming a reliable basis for the subsequent regime-specific explainability analysis.

Figure 8 is here

The regime-specific Grad-CAM visualizations are presented in Fig. 8. Across all three market regimes, macroeconomic variables consistently exhibit the highest activation intensities, indicating their primary importance in ConvFactorNet's factor predictions regardless of market conditions. In contrast, technical indicators persistently remain the least important category in every scenario. While minor differences are observable, for example, market-related features gain slightly more emphasis during the flat regime—these variations do not indicate a fundamental shift in the model's stable macroeconomic-driven decision logic driving the model's predictions. This consistent pattern of feature importance across market regimes underscores ConvFactorNet's robustness in leveraging broadly informative economic indicators. It demonstrates the model's strength in capturing fundamental, long-term market dynamics despite variability in short-term market fluctuations.

Table 7 is here

This consistency is further supported by a quantitative evaluation. Table 7 shows that the relative importance of firms, grouped by market capitalization quartiles, remains stable across all regimes, exhibiting minimal variation among quartiles. This finding reinforces that the model prioritizes macro-level signals over firm-specific characteristics like size.

Table 8 is here

Similarly, the ranking of feature categories, shown in Table 8, remains unchanged across all market states: Macro features are consistently most influential, followed by Market, Financial, and Technical indicators. This consistent hierarchy demonstrates that the model maintains a consistent feature prioritization strategy rooted in macroeconomic dynamics, reflecting a structural rather than a reactive learning approach.

Table 9 is here

To investigate the model's internal behavior more deeply, the top five individual features with the highest Grad-CAM values were analyzed for each market regime, with results summarized in Table 9. In both the bear and flat market regimes, the most influential variables were highly consistent, including the Korea and US Industrial Production Indexes, Gold Futures, WTI Futures, and the USD/KRW exchange rate. This pattern suggests that during periods of economic contraction or uncertainty, the model places greater emphasis on fundamental indicators tied to production, inflation, and currency fluctuations.

In contrast, in bull markets, the dominant features shifted toward fixed-income instruments, particularly short- and medium-term US and Korea Treasury bond yields. This indicates that interest rate expectations and bond market signals play a more central role in the model's factor predictions during market expansions. The emphasis on bond yields suggest ConvFactorNet captures the market's sensitivity to monetary policy and discount rate effects on asset pricing during growth phases. These findings reinforce that the model's predictions are grounded in structural macro-financial variables, with only subtle shifts in emphasis depending on the market regime.

4.2.3. Improving Performance Using Grad-CAM

The previous section confirmed, through Grad-CAM analysis, that the ConvFactorNet model primarily relies on macroeconomic variables for factor prediction. However, it is evident that the importance of each variable varies over time. This observation indicates that the model dynamically adjusts the relevance of input features in response to structural changes in the market. This finding implies that a fixed set of variables may be insufficient to fully capture market dynamic structure, highlighting the necessity for adaptive feature selection mechanisms to enhance predictive performance.

Figure 9 is here

Based on this observation, we conducted an empirical test to determine whether using only the variables identified as important by Grad-CAM at each time point would improve prediction performance. This experiment aims to validate the practical utility of the model interpretability and assess the contribution of a Grad-CAM-based variable selection strategy to real-world investment applications.

The experiment was designed as follows. First, using the ConvFactorNet model trained with the complete set of input variables, Grad-CAM values were calculated at each target date and averaged across all stocks to derive the importance of each variable. Based on the resulting distribution of importance scores, only the variables exceeding a given percentile threshold was selected. The threshold percentiles were adjusted from 10% to 90% in 5% increments to construct various sets of selected variables. Subsequently, new models with the same architecture were retrained at each time point using only these dynamically selected variables to generate predictions and construct a long-short portfolio.

Figure 10 is here

The analysis revealed that models utilizing only the top-ranked variables selected through Grad-CAM generally

outperformed the baseline model. In particular, when variables within the top 40% to 70% percentile range were used, the models achieved the highest cumulative returns and Sharpe Ratios, suggesting that this range allows for enhanced informational density without excessive variable reduction. These models also recorded slightly higher Tertile Accuracy compared to the baseline, indicating improved precision in ranking stock returns. In contrast, when the model was trained using only the top 10% of variables, performance declined significantly due to a loss of critical information.

Figure 11 is here

Table 10 provides a detailed comparison of investment performance for the best-performing percentile model (top 45%), the worst-performing (top 10%), and the baseline model that used all available features. The model utilizing only the top 45% of features achieved approximately 65% higher cumulative returns compared to the baseline, along with higher Tertile Accuracy and a lower Maximum Drawdown. This result suggests that dynamic feature selection guided by Grad-CAM extends beyond model interpretability and directly improves investment profitability. The widening performance gap over time is visualized in Fig. 12.

Figure 12 is here

Table 10 is here

Additionally, an experiment was conducted to evaluate the temporal validity of feature importance by analyzing the effect of averaging Grad-CAM values over various time windows. The detailed results of this analysis are provided in [Appendix D](#). These findings demonstrate that feature importance derived from Grad-CAM is not merely a tool for model interpretation but can serve as a practical mechanism for enhancing predictive performance and improving investment outcomes. Specifically, the strategy of dynamically selecting relevant input features at each time point proves to be both feasible and effective. Compared to conventional approaches that rely on fixed input feature sets, this dynamic selection framework offers superior adaptability to changing market conditions and improved profitability.

5. Conclusion

This study proposes the Convolutional Factor Network (ConvFactorNet), a deep learning-based factor prediction model and presents a novel methodology that links factor prediction to return forecasting and practical investment strategies in financial markets. Unlike traditional factor models, which rely on predefined static indicators (e.g., PER, SMB, HML) and are constrained by linear regression approaches (e.g., OLS), ConvFactorNet captures dynamic market changes and complex interactions among variables through a data-driven neural network architecture. Furthermore, by incorporating Grad-CAM, the model ensures interpretability, with explanations quantitatively validated for their positive impact on both predictive accuracy and investment performance.

The contributions of this study are threefold. First, we propose ConvFactorNet, a CNN-based architecture that automatically extracts dynamic and nonlinear factors from raw financial data. Second, we link factor predictions to a long-short market-neutral strategy and show that ConvFactorNet outperforms benchmark strategies in the Korean stock market with higher cumulative returns and Sharpe ratios. Third, we integrate Grad-CAM for interpretability and demonstrate that timely variable feature selection guided by its importance scores significantly enhances investment performance, with models using the top 40–70% of variables substantially outperforming those relying on the full set of variables.

In summary, this study shows that a deep learning-based factor model can achieve both high predictive power and interpretability. The combination of ConvFactorNet and Grad-CAM offers a scalable and transparent framework for adaptive variable selection and investment strategy design, providing a practical pathway for applying interpretable artificial intelligence to real-world financial decision-making. These findings suggest that combining deep learning with explainable AI offers a powerful framework for advancing financial forecasting and decision support systems with both accuracy and transparency.

Future Work

This study developed a deep learning-based factor prediction model using Korean stock market data and empirically validated its practical effectiveness in investment strategies. Future research should expand the scope to global

financial markets to assess the generalizability and scalability of the proposed model. Since economic structures and financial systems differ significantly across countries, comparative evaluations using data from the United States, Europe, and emerging markets would be particularly valuable. Such analyses would allow systematic exploration of optimal hyperparameter settings and input configurations tailored to region-specific market characteristics, thereby verifying universality of the proposed model as an investment strategy.

Another promising direction is to further enhance model interpretability beyond visual analyses. While this study applied Grad-CAM to visually analyze variable importance across time, future work could establish a statistical framework that integrates interpretability with quantitative evaluation metrics to more precisely measure variable contributions. Moreover, linking interpretability with domain expert judgment may facilitate the development of semi-automated strategy design tools or adaptive factor adjustment systems, thereby improving the transparency and credibility of investment decisions.

6. References

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7. Figures

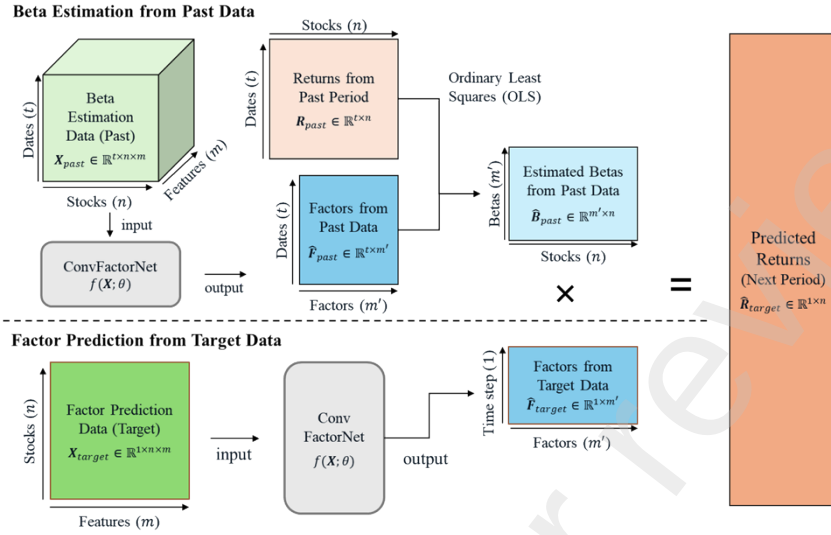


Fig 1. Overview of the Predictive Process

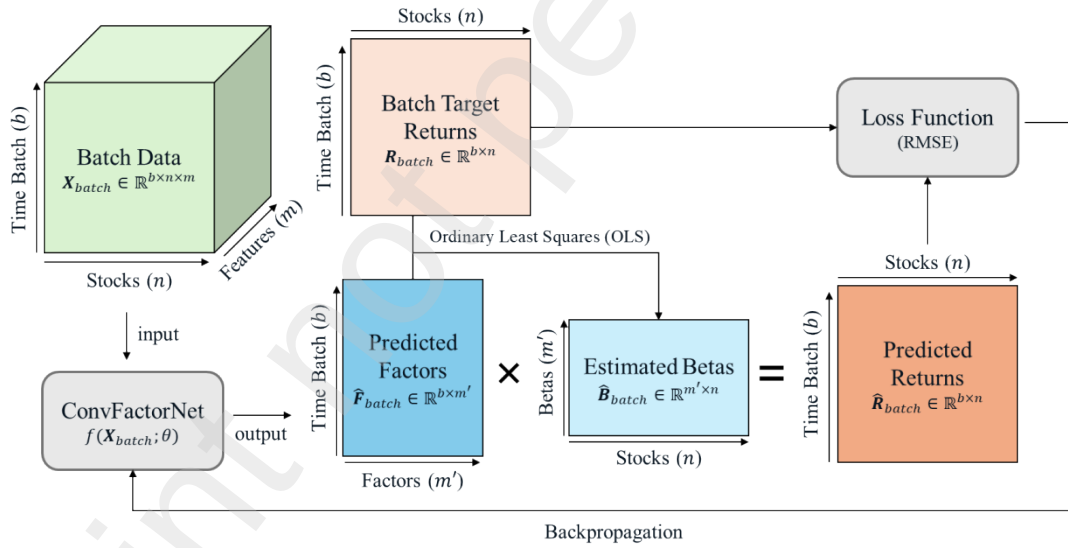


Fig 2. Overview of the Model Training Process

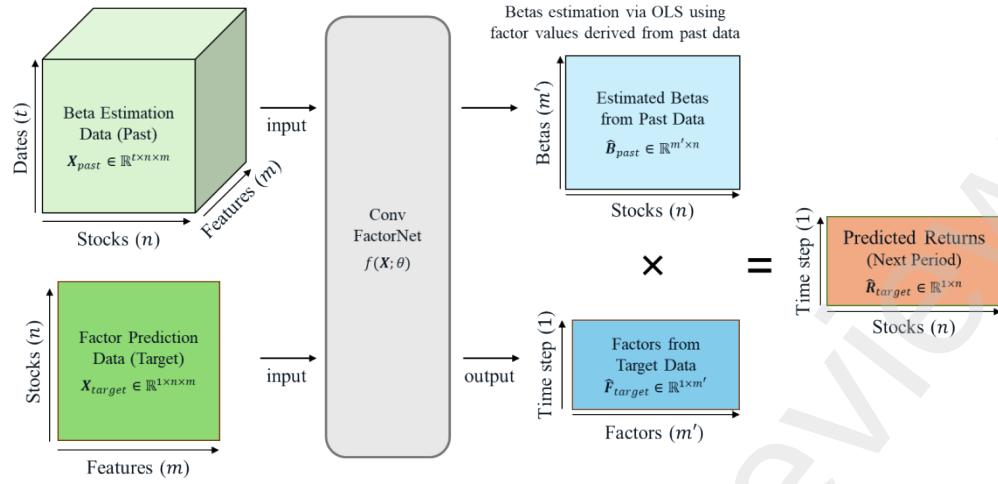


Fig 3. Overview of the Target Date Prediction Process

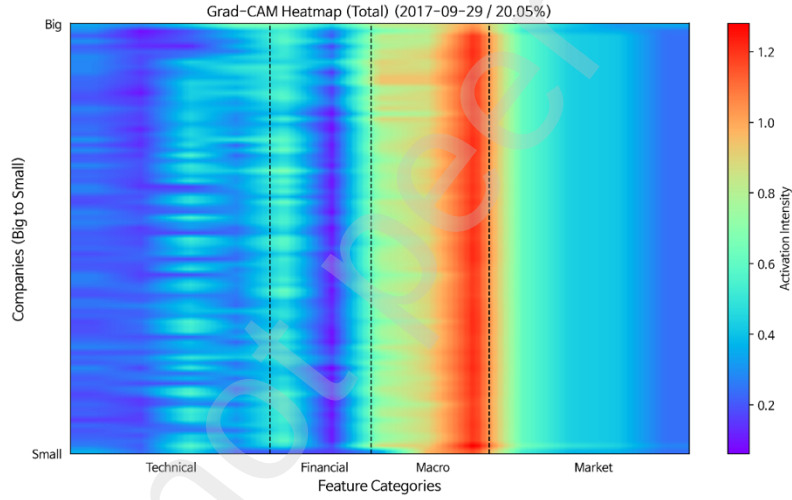


Fig 4. Example of Feature Importance Map Generated Using Grad-CAM

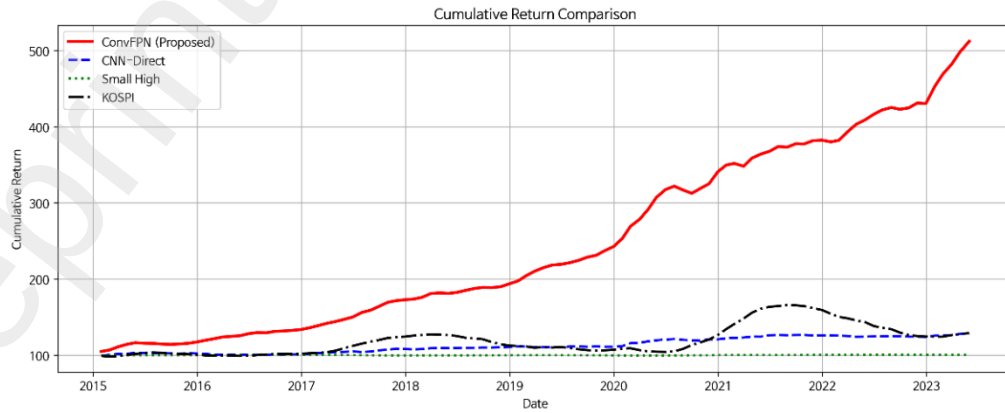


Fig 5. Cumulative Return Comparison of the Proposed Model and Benchmark Strategies

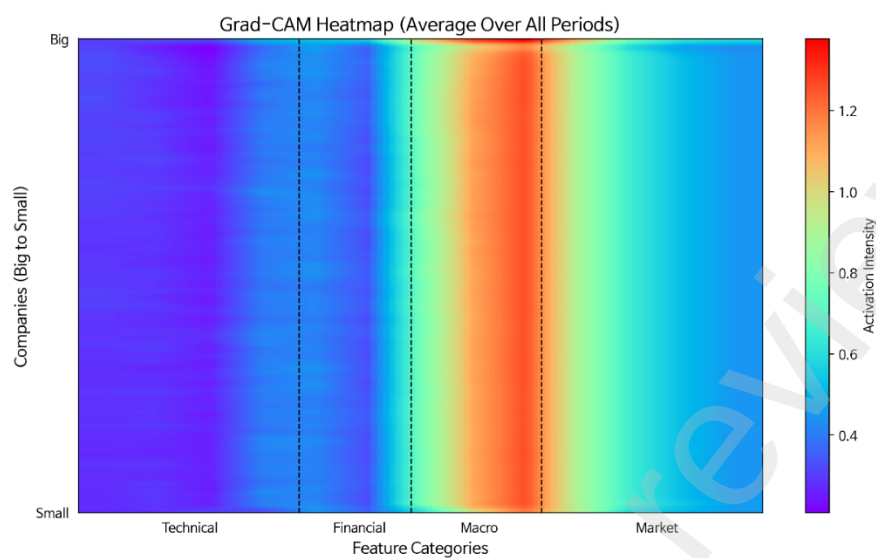


Fig 6. Average Grad-CAM Visualization Over the Target Period (2015-01-30 ~ 2023-05-31)

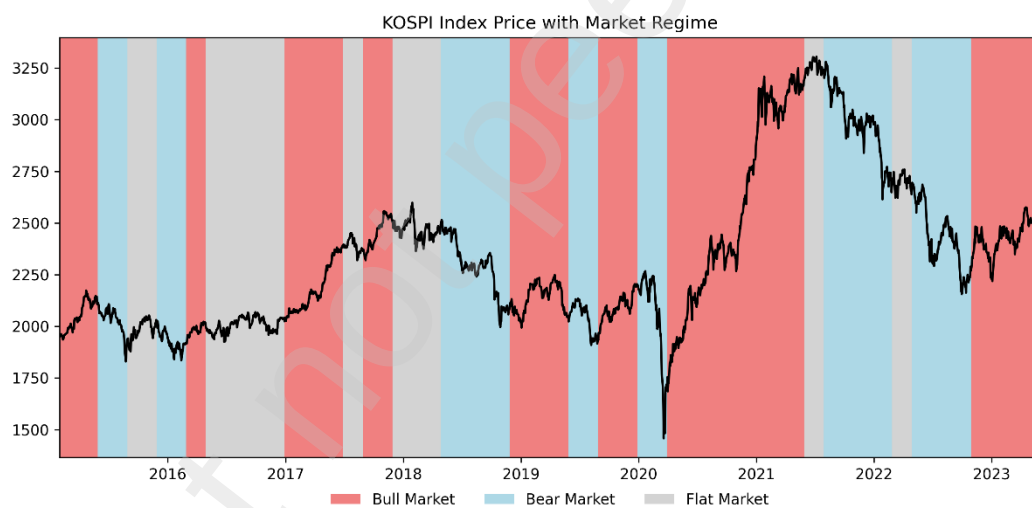


Fig 7. KOSPI Index Close Price with Bull, Bear, and Flat Market Regimes

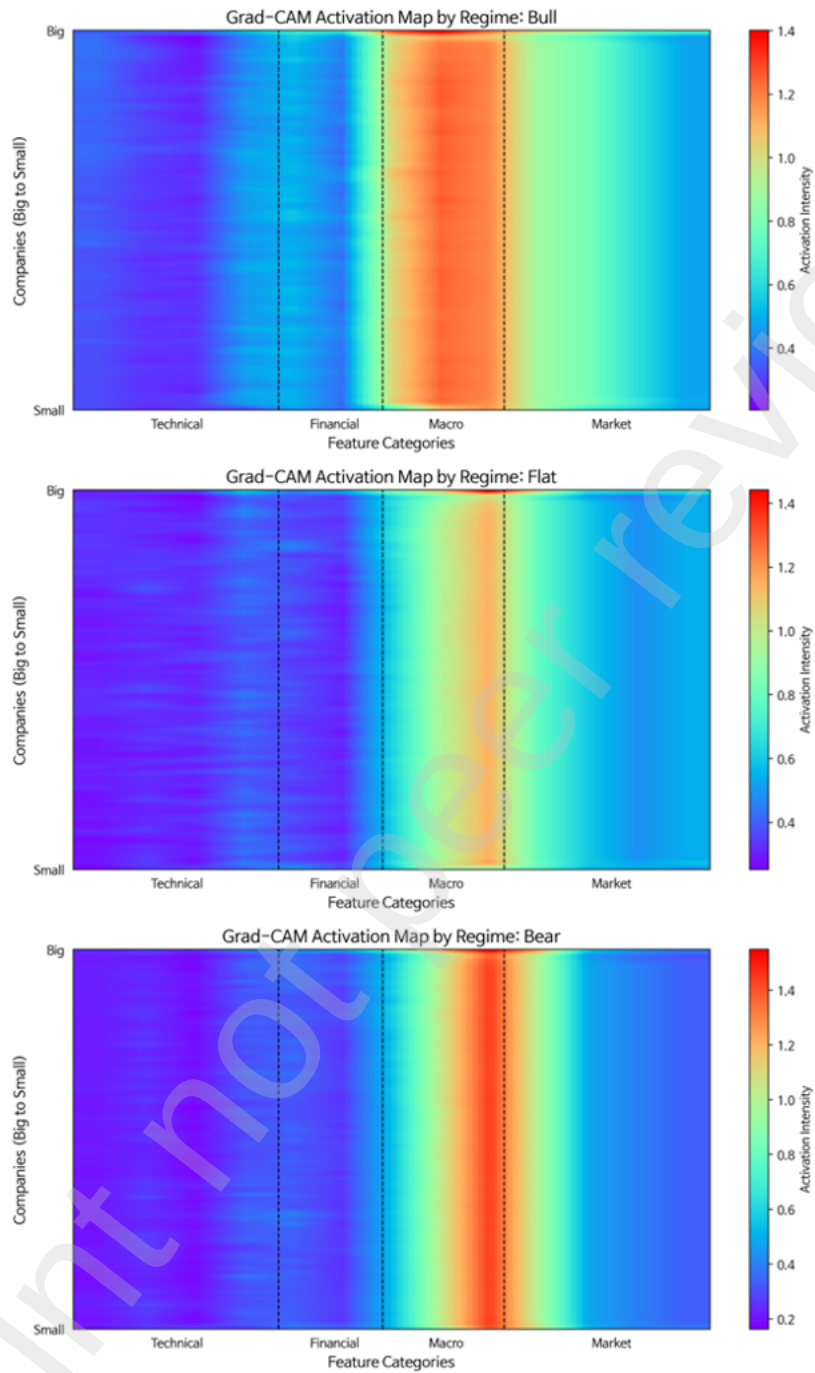


Fig 8. Grad-CAM Visualization Across Market Regimes

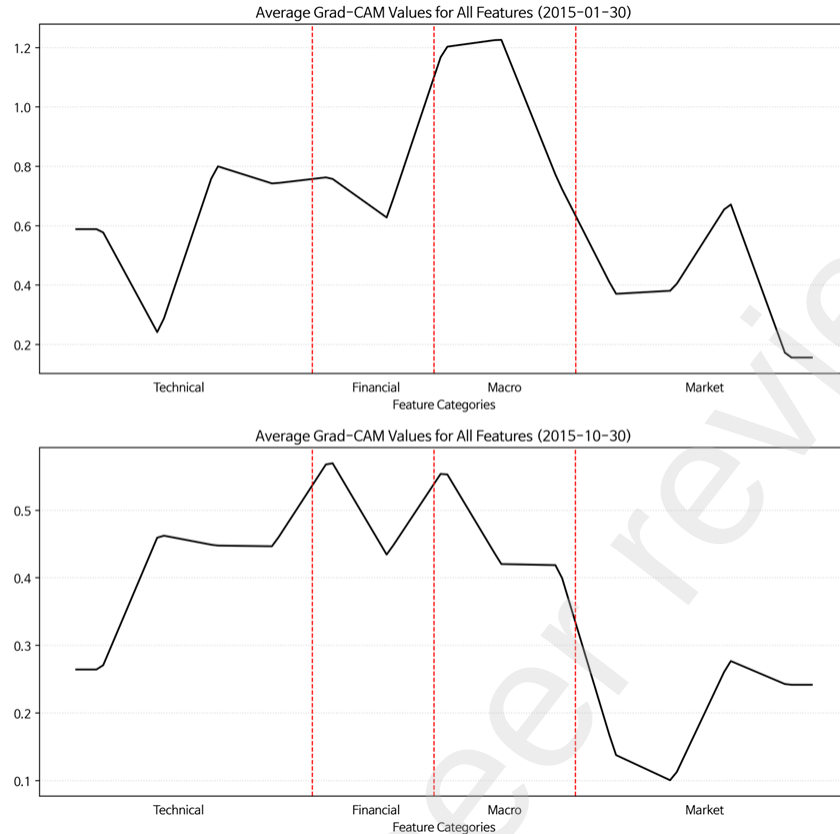


Fig 9. Average Grad-CAM Values on the Target Date

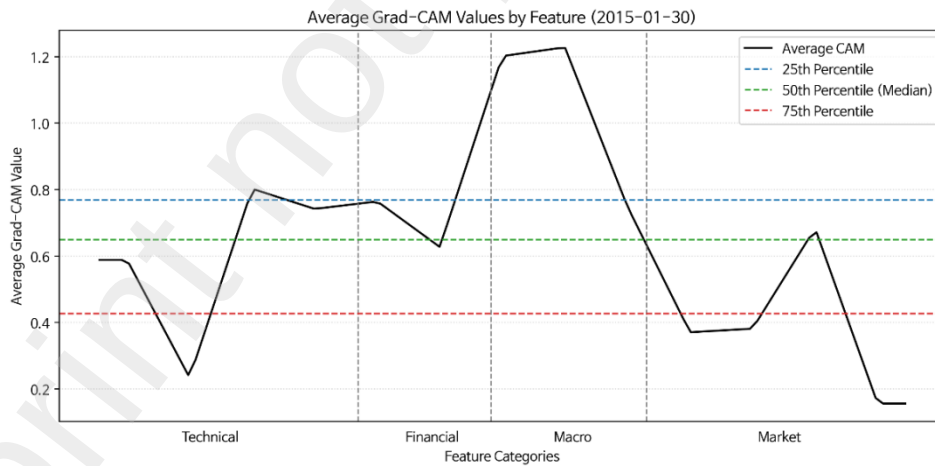


Fig 10. Average Grad-CAM Values per Feature on 2015-01-30, with Percentile Thresholds

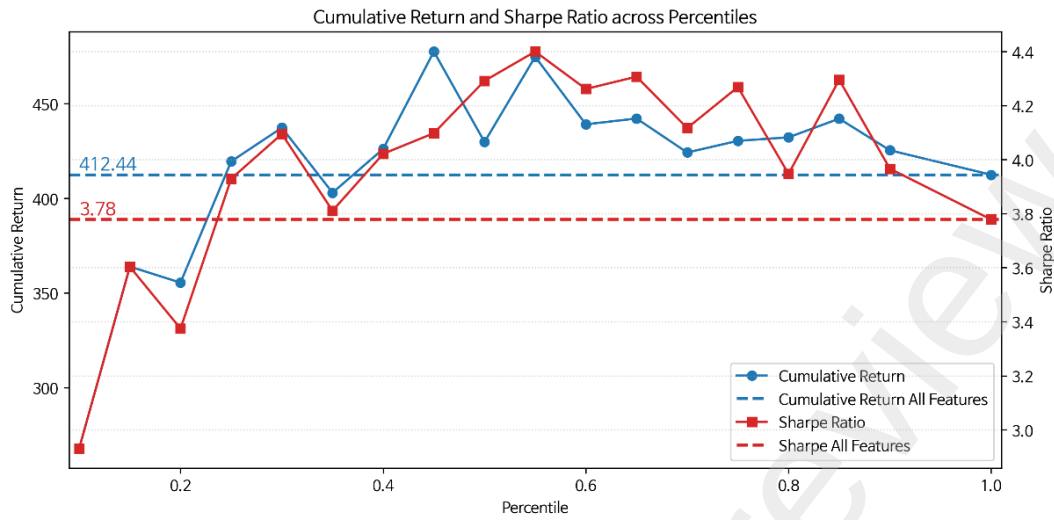


Fig 11. Cumulative Return and Sharpe Ratio across Percentiles

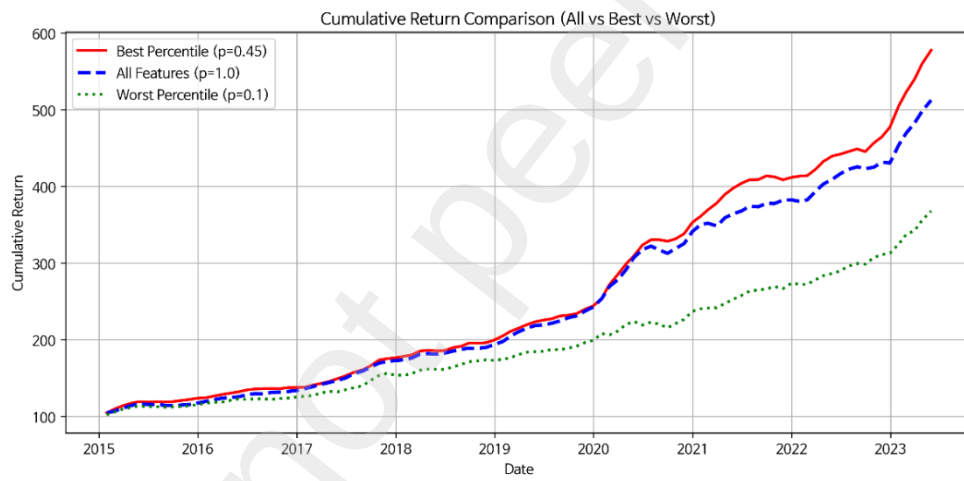


Fig 12. Cumulative Returns: All Features vs. Best and Worst Percentiles

8. Tables

Table 1. CNN Model Parameters for Input Data of Shape (Batch Size, n, m)

Layer	Input shape	Output shape	Number of parameters
Conv2d-1	(b, 1, n, m)	(b, 16, n, m)	160
BatchNorm2d-2	(b, 16, n, m)	(b, 16, n, m)	32
ReLU-3	(b, 16, n, m)	(b, 16, n, m)	-
MaxPool2d-4	(b, 16, n, m)	(b, 16, n//2, m//2)	-
Conv2d-5	(b, 16, n//2, m//2)	(b, 32, n//2, m//2)	4,640
BatchNorm2d-6	(b, 32, n//2, m//2)	(b, 32, n//2, m//2)	64
ReLU-7	(b, 32, n//2, m//2)	(b, 32, n//2, m//2)	-
MaxPool2d-8	(b, 32, n//2, m//2)	(b, 32, n//4, m//4)	-
Conv2d-9	(b, 32, n//4, m//4)	(b, 64, n//4, m//4)	18,496
BatchNorm2d-10	(b, 64, n//4, m//4)	(b, 64, n//4, m//4)	128
ReLU-11	(b, 64, n//4, m//4)	(b, 64, n//4, m//4)	-
MaxPool2d-12	(b, 64, n//4, m//4)	(b, 64, n//8, m//8)	-
Conv2d-13	(b, 64, n//8, m//8)	(b, m', n//8, m//8)	5,770
Mean-14 (GAP)	(b, m', n//8, m//8)	(b, m')	-

Table 2. Predictive Performance Rankings by Accuracy under Hyperparameter Settings

Rank	Lookback Period	Target Days	m'	Accuracy (%)	F1 Score (Macro)	Rank	Lookback Period	Target Days	m'	Accuracy (%)	F1 Score (Macro)
1	2	120	20	54.72	0.5002	11	3	20	5	53.23	0.4529
2	2	120	10	54.72	0.4949	12	3	60	5	53.20	0.4651
3	3	120	20	54.52	0.4978	13	3	120	5	53.07	0.4780
4	1	120	10	54.44	0.5002	14	2	60	20	52.95	0.4963
5	1	120	20	54.36	0.5019	15	1	60	10	52.87	0.4873
6	1	60	20	54.07	0.5028	16	2	60	10	52.54	0.4860
7	3	60	20	53.58	0.4935	17	2	120	5	52.48	0.4729
8	1	120	5	53.43	0.4900	18	1	20	10	52.31	0.4741
9	3	60	10	53.37	0.4845	19	1	20	20	52.04	0.4858
10	3	120	10	53.30	0.4892	20	3	20	10	52.00	0.4657

Table 3. Investment Performance Rankings by Sharpe Ratio under Hyperparameter Settings

Rank	Lookback Period	Target Days	m'	Tertile Acc (%)	Cum. Return (%)	CAGR (%)	Volatility (%)	Sharpe	MDD (%)
1	1	120	10	38.52	412.44	21.67	5.21	3.78	-2.90
2	1	120	20	38.33	330.93	19.16	4.65	3.69	-1.89
3	1	60	20	37.71	437.83	22.38	6.22	3.27	-3.82
4	2	120	20	37.97	256.08	16.47	4.45	3.25	-2.66
5	1	120	5	37.20	260.09	16.62	4.80	3.04	-4.81
6	2	120	10	37.41	229.27	15.38	4.62	2.89	-3.90
7	3	120	20	37.54	178.28	13.07	3.86	2.87	-2.42
8	2	60	20	36.88	297.67	18.02	5.98	2.68	-3.95
9	1	20	20	36.53	534.76	24.84	9.79	2.33	-8.02
10	1	60	10	36.71	275.66	17.22	7.00	2.17	-5.66

Table 4. Average Grad-CAM Values by Feature Category During Target Period

Category	Average Grad-CAM Value
Macro	1.0478
Market	0.6670
Financial	0.4314
Technical	0.3195

Table 5. Top 10 Features by Grad-CAM Values During Target Period

Rank	Feature Name	Grad-CAM Value
1	Korea Industrial Production Index	1.2542
2	US Industrial Production Index	1.2366
3	Gold Futures	1.2365
4	WTI Futures	1.2187
5	US Treasury Bond 10 Years	1.201
6	USD/KRW Exchange Rate US	1.1931
7	Treasury Bond 5 Years	1.1832
8	US Treasury Bond 1 Year	1.1654
9	KOSPI Return 1 Day	1.1496
10	US Treasury Bond 6 Months	1.1477

Table 6. Frequency of Market Regimes

Market Regime	Frequency (Months)
Bull	46
Flat	23
Bear	32

Table 7. Grad-CAM Rankings by Market Capitalization Across Market Regimes

Regime	Rank 1	Rank 2	Rank 3	Rank 4
Bull	Q1	Q2	Q3	Q4
Bear	Q1	Q2	Q3	Q4
Flat	Q1	Q2	Q3	Q4

Table 8. Grad-CAM Rankings by Feature Category Across Market Regimes

Regime	Rank 1	Rank 2	Rank 3	Rank 4
Bull	Macro	Market	Financial	Technical
Bear	Macro	Market	Financial	Technical
Flat	Macro	Market	Financial	Technical

Table 9. Top 5 Features with Highest Grad-CAM Values Across Market Regimes

Regime	Bull	Bear	Flat
Rank 1	US Treasury Bond 1 Month	Korea Industrial Production Index	Korea Industrial Production Index
Rank 2	US Treasury Bond 3 Months	US Industrial Production Index	US Industrial Production Index
Rank 3	US Treasury Bond 6 Months	Gold Futures	Gold Futures
Rank 4	US Treasury Bond 1 Year	USD/KRW Exchange Rate	WTI Futures
Rank 5	Korea Treasury Bond 10 Years	WTI Futures	USD/KRW Exchange Rate

Table 10. Investment Performance: All Features vs. Best and Worst Percentiles

Model Type	Tertile Acc (%)	Cum. Return (%)	CAGR (%)	Volatility (%)	Sharpe	MDD (%)
All Features (p=1.0)	38.52	412.44	21.669	5.206	3.778	-2.903
Best Percentile (p=0.45)	38.91	477.51	23.427	5.229	4.098	-1.224
Worst Percentile (p=0.1)	37.99	267.90	16.925	5.091	2.931	-3.682

9. Algorithm

Algorithm 1. Market Regime Classification Based on Monthly Returns

Require: Daily closing prices P_t , window size w , threshold θ
 Ensure: Market regime labels $Regime_t \in \{bull, bear, flat\}$

- 1: Resample P_t to monthly closing prices C_t using month-end values
- 2: Compute monthly returns: $R_t \leftarrow \frac{C_t - C_{t-1}}{C_{t-1}}$
- 3: Compute centered moving average of returns:
- 4: for each month t do
- 5: $MA_t \leftarrow \frac{1}{w} \sum_{i=-k}^k R_{t+i}$ where $k = \lfloor \frac{w}{2} \rfloor$
- 6: end for
- 7: Initial regime labeling:
- 8: for each month t do
- 9: if $MA_t > \theta$ then
- 10: $Label_t \leftarrow bull$
- 11: else if $MA_t < -\theta$ then
- 12: $Label_t \leftarrow bear$
- 13: else
- 14: $Label_t \leftarrow flat$
- 15: end if
- 16: end for
- 17: Label smoothing:
- 18: for $t = 1$ to $T - 2$ do
- 19: if $Label_t \neq Label_{t-1}$ and $Label_t \neq Label_{t+1}$ then
- 20: $Label_t \leftarrow Label_{t-1}$
- 21: end if
- 22: end for
- 23: return $Regime_t \leftarrow Label_t$

10. Appendix

We include several appendices with additional details, analyses, and supplementary results. In Appendix A, we provide a detailed description of the 110 input features and the data preprocessing methodology. Appendix B offers a detailed breakdown of the model's investment performance, including a full hyperparameter tuning analysis, diagnostic classification results, and a comprehensive comparison against benchmark strategies. In Appendix C, we extend our explainability analysis by investigating the consistency of feature importance patterns across different industry sectors. Finally, Appendix D details an additional experiment that evaluates the temporal validity of feature importance by analyzing the effect of averaging Grad-CAM scores over various time windows.

Appendix A: Dataset and Preprocessing

The study utilizes a dataset of daily observations from January 2010 to December 2023, collected via DataGuide. To facilitate a comprehensive market analysis, we selected 110 features, which were organized into four distinct categories: Technical, Financial, Macro, and Market Indicators.

Technical Indicators include a variety of technical analysis metrics derived from stock price and volume data, computed using Python's TA-Lib library. During the data preprocessing phase, feature-specific scaling was applied to address outliers. For features with excessively large or small values, a logarithmic transformation was used to normalize the data distribution. Price-dependent indicators (e.g., moving averages, Bollinger Bands) were normalized by dividing by the closing price, while volume-dependent indicators (e.g., moving average volume, On-Balance Volume (OBV), Accumulation/Distribution (AD)) were scaled by the trading volume.

Financial Indicators were selected from key items in firms' financial statements, as they play a critical role in stock valuation and the assessment of corporate growth. To accurately reflect real market conditions, the reporting lag of financial statements was carefully considered. To prevent look-ahead bias, financial data were included only after a four-month delay from the official disclosure date, ensuring that the information was publicly available to market participants at the time of model training. For instance, the 2023 financial data for firms with a fiscal year ending in December were incorporated into the beta estimation dataset starting from April 2024. This approach ensures a realistic investment simulation by preventing the model from accessing future information.

Macro Indicators represent key macroeconomic variables included to capture overall market trends and external economic conditions. As these variables are typically provided on a monthly basis, they were transformed into daily data using a forward-fill technique. This method replaces missing daily values with the most recent available monthly data, ensuring that the model does not inadvertently access future information during training and maintains alignment with real-world forecasting scenarios.

Market Indicators were included to enable the model to learn the relationship between factors and broader market movements. In this study, historical returns of major market indices—such as the KOSPI, KOSDAQ, NASDAQ, and S&P 500—were used. To capture short-, medium-, and long-term market trends, returns over various horizons were incorporated: 1-day, 1-week, 1-month, 3-month, 6-month, and 12-month intervals.

Table 11. Feature Names and Categories Used in Experiments

Name	Category	Name	Category
Market Capitalization	Technical	M1	Macro
Return 1 Day	Technical	M2	Macro
Return 5 Days	Technical	Lf	Macro
Return 20 Days	Technical	CD91	Macro
Return 50 Days	Technical	Korea Treasury Bond 1 Year	Macro
Return 100 Days	Technical	Korea Treasury Bond 3 Years	Macro
Return 200 Days	Technical	Korea Treasury Bond 5 Years	Macro
MA20	Technical	Korea Treasury Bond 10 Years	Macro
MA50	Technical	US Treasury Bond 1 Month	Macro
MA100	Technical	US Treasury Bond 3 Months	Macro
MA200	Technical	US Treasury Bond 6 Months	Macro
EMA12	Technical	US Treasury Bond 1 Year	Macro
EMA26	Technical	US Treasury Bond 5 Years	Macro
MAV20	Technical	US Treasury Bond 10 Years	Macro
MAV50	Technical	WTI Futures	Macro
MAV100	Technical	Gold Futures	Macro
MAV200	Technical	Korea Industrial Production Index	Macro
BBANDS Upper	Technical	US Industrial Production Index	Macro
BBANDS Middle	Technical	USD/KRW Exchange Rate	Macro
BBANDS Lower	Technical	KOSPI Return 1 Day	Market
RSI	Technical	KOSPI Return 1 Week	Market
MACD	Technical	KOSPI Return 1 Month	Market
MACD Signal	Technical	KOSPI Return 3 Months	Market
MACD Histogram	Technical	KOSPI Return 6 Months	Market
Stochastic %K	Technical	KOSPI Return 12 Months	Market
Stochastic %D	Technical	KOSPI 200 Return 1 Day	Market
ADX	Technical	KOSPI 200 Return 1 Week	Market
CCI	Technical	KOSPI 200 Return 1 Month	Market
OBV	Technical	KOSPI 200 Return 3 Months	Market
AD	Technical	KOSPI 200 Return 6 Months	Market
Williams %R	Technical	KOSPI 200 Return 12 Months	Market
BOP	Technical	KOSDAQ Return 1 Day	Market
ATR	Technical	KOSDAQ Return 1 Week	Market
NATR	Technical	KOSDAQ Return 1 Month	Market
True Range	Technical	KOSDAQ Return 3 Months	Market
Total Assets	Financial	KOSDAQ Return 6 Months	Market
Total Liabilities	Financial	KOSDAQ Return 12 Months	Market
Total Equity	Financial	NASDAQ Composite Return	Market
Sales Revenue	Financial	NASDAQ Return 1 Week	Market
Operating Profit	Financial	NASDAQ Return 1 Month	Market

Name	Category	Name	Category
Net Income	Financial	NASDAQ Return 3 Months	Market
Debt Ratio	Financial	NASDAQ Return 6 Months	Market
Reserve Ratio	Financial	NASDAQ Return 12 Months	Market
Operating Profit Margin	Financial	S&P 500 Return 1 Day	Market
Net Profit Margin	Financial	S&P 500 Return 1 Week	Market
ROA	Financial	S&P 500 Return 1 Month	Market
ROE	Financial	S&P 500 Return 3 Months	Market
EPS	Financial	S&P 500 Return 6 Months	Market
BPS	Financial	S&P 500 Return 12 Months	Market
EBITDA	Financial	NYSE Return 1 Day	Market
Dividend	Financial	NYSE Return 1 Week	Market
PER	Financial	NYSE Return 1 Month	Market
PBR	Financial	NYSE Return 3 Months	Market
Consumer Price Index	Macro	NYSE Return 6 Months	Market
Producer Price Index	Macro	NYSE Return 12 Months	Market

Appendix B: Detailed Investment Performance Analysis

This appendix provides a detailed examination of the model's investment performance, focusing on the impact of hyperparameter tuning, diagnostic classification analysis, and a comprehensive comparison against benchmark strategies.

B.1. Diagnostic Classification Analysis

To understand the drivers of performance, we conducted a diagnostic analysis of the models' classification abilities. The confusion matrices for the best- and worst-performing models reveal significant differences. For the binary (Up/Down) task, the best model showed statistically meaningful directional accuracy, with a particular strength in predicting downward movements. For the more granular tertile (Low, Medium, High) classification, the best model's confusion matrix exhibited a strong diagonal alignment, confirming its effectiveness in correctly ranking stocks by future returns. In contrast, the worst model's predictions were dispersed, indicating performance comparable to random guessing.

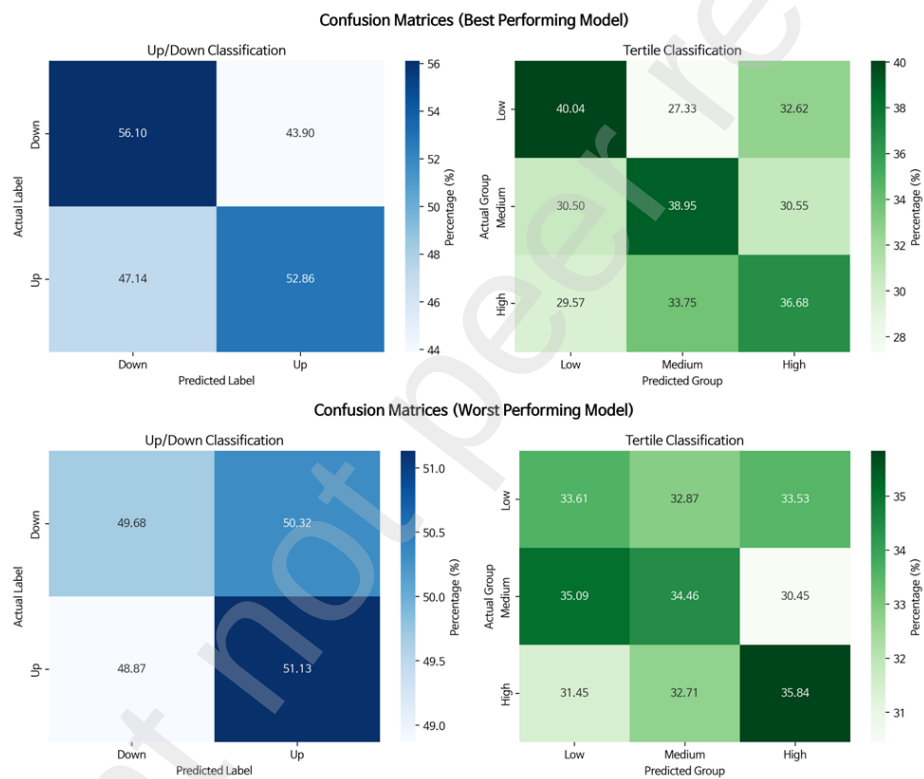


Fig 13. Confusion Matrices for Up/Down and Tertile Classification in Best vs. Worst Models

This ranking ability is further quantified by analyzing the average monthly returns of the predicted stock groups (terciles). As shown in Table 12, the best-performing model generated a substantial return spread of 1.64% per month between its High and Low tercile groups. This large spread is the source of the long-short strategy's profitability. The worst-performing model, however, produced a negligible spread of only 0.19%, rendering the strategy ineffective. These results confirm that the model's investment success is underpinned by its precise cross-sectional ranking of stocks.

Table 12. Average Monthly Returns of Stocks in Each Group

Rank	Lookback Period	Target Days	m	Low (%)	Medium (%)	High (%)	High - Low (%)
1	1	120	10	-0.329	0.289	1.313	1.642
2	1	120	20	-0.289	0.385	1.177	1.466
3	1	60	20	-0.369	0.268	1.326	1.695
4	2	120	20	-0.178	0.334	1.095	1.273
5	1	120	5	-0.135	0.257	1.151	1.286
6	2	120	10	-0.157	0.369	1.039	1.196
7	3	120	20	-0.059	0.302	0.965	1.024
8	2	60	20	-0.244	0.349	1.147	1.391
9	1	20	20	-0.513	0.341	1.372	1.885
10	1	60	10	-0.241	0.369	1.098	1.339
11	3	120	5	0.044	0.327	0.838	0.794
12	2	120	5	0.039	0.423	0.790	0.751
13	3	120	10	0.049	0.302	0.857	0.808
14	3	60	20	-0.127	0.441	0.908	1.035
15	2	20	20	-0.296	0.444	1.073	1.370
16	3	20	10	-0.285	0.470	1.087	1.372
17	2	60	10	-0.002	0.411	0.844	0.846
18	1	20	10	-0.306	0.419	1.087	1.393
19	2	20	10	-0.234	0.387	1.067	1.301
20	2	20	5	-0.240	0.546	0.915	1.155
21	1	60	5	-0.005	0.446	0.785	0.790
22	1	20	5	-0.138	0.355	0.984	1.122
23	3	60	10	0.079	0.389	0.754	0.675
24	2	120	40	0.218	0.327	0.707	0.489
25	3	60	5	0.047	0.474	0.701	0.654
26	2	60	5	0.067	0.462	0.724	0.658
27	1	120	40	0.224	0.322	0.728	0.505
28	3	20	20	0.051	0.416	0.807	0.755
29	3	20	5	-0.064	0.523	0.815	0.879
30	1	20	40	0.054	0.410	0.737	0.683
31	1	60	40	0.236	0.335	0.656	0.419
32	3	60	40	0.229	0.352	0.640	0.411
33	2	20	40	0.190	0.392	0.640	0.450
34	3	120	40	0.321	0.296	0.592	0.272
35	3	20	40	0.325	0.404	0.545	0.220
36	2	60	40	0.297	0.471	0.485	0.188

B.2. Benchmark Strategy Comparison

To contextualize ConvFactorNet's performance, it was evaluated against three distinct benchmark strategies, which are defined in Table 13.

Table 13. Benchmark Strategies for Investment Performance Evaluation

Benchmark	Description
KOSPI Index	A representative stock index of the Korean equity market. Used to evaluate whether the model generates excess returns over the market benchmark.
CNN Direct	A model designed to directly predict stock returns by modifying the final layer of ConvFactorNet, used to verify the effectiveness of predicting factors instead of returns.
Small High Strategy	A long-short strategy that buys undervalued small-cap stocks and sells overvalued large-cap stocks, based on market capitalization and price-to-book ratio (PBR).

The empirical results, summarized in Table 14, demonstrate that the ConvFactorNet strategy substantially outperformed all benchmarks. Notably, it surpassed the CNN-Direct model (Sharpe Ratio of 3.778 vs. 0.474), which predicts returns directly without the factor-learning step. This highlights the structural advantage of the proposed factor-based approach. Furthermore, ConvFactorNet's Tertile Accuracy (38.52%) was significantly higher than that of CNN-Direct (33.78%), confirming that the factor structure enhances the model's ability to accurately rank stocks. The strategy also generated significant excess returns over the KOSPI index and proved far superior to a traditional Small High factor strategy, which failed to produce meaningful returns during the evaluation period.

Table 14. Investment Performance Metrics (ConvFactorNet and Benchmarks)

Strategy	Tertile Acc (%)	Cum. Return (%)	CAGR (%)	Volatility (%)	Sharpe	MDD (%)
ConvFactorNet	38.52	412.44	21.669	5.206	3.778	-2.903
KOSPI	-	29.16	3.120	6.709	0.167	-25.138
CNN-Direct	33.78	28.84	3.089	2.299	0.474	-2.737
Small High	-	0.33	0.040	0.103	-18.983	-0.561

Appendix C: Explainability Analysis by Industry Sector

To validate the robustness of the model's decision-making process, we conducted a Grad-CAM analysis to determine if feature importance patterns remained consistent across different industry sectors. The objective was to assess whether ConvFactorNet prioritizes different inputs based on a firm's industry.

We classified all firms into ten sectors based on the FnGuide Industry Classification Standard (FICS), as detailed in Table 15. For each evaluation date, stock-level Grad-CAM values were computed and then averaged within their respective sectors. These sector-level averages were then aggregated over the entire test period to identify stable patterns.

Table 15. Sector Classification and Weight based on FICS

Sector	Weight (%)
IT	22.38
Consumer Discretionary	18.97
Industrials	16.25
Health Care	16.09
Materials	12.25
Consumer Staples	7.63
Energy	2.54
Financials	2.06
Utilities	1.16
Communication Services	0.68

The analysis revealed a striking consistency in feature importance across all ten industry sectors. As detailed in Table 16, macroeconomic indicators consistently registered the highest Grad-CAM values, followed by market, financial, and technical indicators. This uniform ranking suggests that the model's factor estimation is driven primarily by systemic economic conditions rather than idiosyncratic, sector-specific dynamics.

Table 16. Average Grad-CAM Values by Sector During the Test Period

Sector	Financial	Macro	Market	Technical
IT	0.43	1.05	0.67	0.32
Consumer Discretionary	0.43	1.05	0.67	0.32
Financials	0.44	1.04	0.66	0.32
Industrials	0.43	1.05	0.67	0.32
Materials	0.43	1.05	0.67	0.32
Energy	0.43	1.04	0.66	0.32
Utilities	0.44	1.05	0.67	0.32
Health Care	0.43	1.05	0.67	0.32
Communication Services	0.45	1.04	0.66	0.31
Consumer Staples	0.43	1.05	0.67	0.32

The results imply that ConvFactorNet operates on the premise that all sectors are fundamentally anchored to the same macroeconomic information structure. This reinforces the model's systematic orientation toward capturing broad, long-term economic trends over industry-level variations.

Appendix D: Temporal Validity of Grad-CAM Feature Importance

To evaluate the temporal validity of feature importance derived from Grad-CAM, we conducted an additional experiment. The objective was to determine whether averaging importance scores over a historical window would improve performance compared to using only the most recent, point-in-time scores. We tested four rolling average windows: 1, 2, 3, and 6 months. For each time point, we selected the top 45% of features based on their averaged Grad-CAM scores from the corresponding window, retrained the model, and evaluated the resulting investment performance.

The results, presented in Figure 14 and Table 17, indicate that using averaged feature importance scores does not enhance model performance. The model utilizing the most recent Grad-CAM values consistently outperformed all models that used averaged importance scores across the various window sizes. This finding suggests that the relevance of input features is highly time-sensitive and diminishes rapidly. Relying on the most up-to-date feature importance is therefore critical for effectively capturing evolving market dynamics.

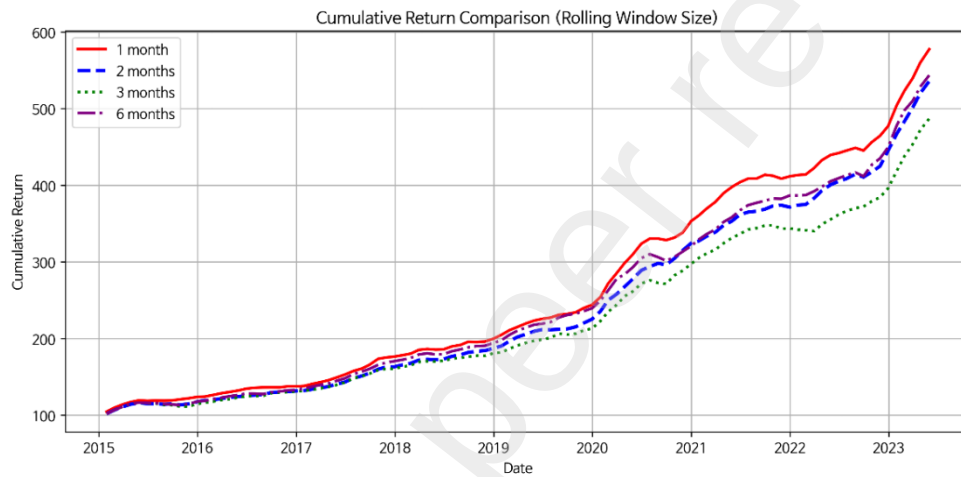


Fig 14. Cumulative Returns by Rolling Window Size (1M, 2M, 3M, 6M)

Table 17. Investment Performance Metrics by Rolling Window Size

Window Size	Tertile Acc (%)	Cum. Return (%)	CAGR (%)	Volatility (%)	Sharpe	MDD (%)
1 month	38.91	477.51	23.427	5.229	4.098	-1.224
2 months	38.86	435.88	22.324	4.943	4.112	-2.262
3 months	38.66	387.07	20.930	5.335	3.549	-4.446
6 months	39.09	443.54	22.532	5.093	4.032	-2.797

Ultimately, this analysis reinforces that feature importance derived from Grad-CAM is not merely an interpretive tool but a practical mechanism for enhancing investment outcomes. The strategy of dynamically selecting the most relevant input features at each specific time point proves to be both feasible and effective. This dynamic framework offers superior adaptability compared to conventional approaches that rely on fixed feature sets, providing a robust solution that satisfies the dual demands for model interpretability and real-world applicability.